Qualification Pack

Technician Railway Track Electrification

QP Code: PSS/Q2501

Version: 1.0

NSQF Level: 4

Power Sector Skill Council || Plot No. 4, Institutional Area, CBIP Building, Malcha Marg, Chanakyapuri New Delhi-110021 || email:akbhagatin@yahoo.co.in

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PSS/Q2501: Technician Railway Track Electrification

Brief Job Description

The individual in the job will perform various activities related to railway track electrification such as testing of soil, erection of mast, stringing of overhead wires, testing & inspection of the other OHE (Overhead Equipment) etc. in adherence to procedures and standards mentioned in Research Design and Standard Organization (RDSO). The individual also performs erection & commissioning of railway substations.

Personal Attributes

The job requires the individual to have good physical strength and ability to work for long working hours/nights, good eye visibility and ability to communicate. The individual should be ethical and well behaved.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. PSS/N1331: Apply basic health and safety practices for power related work
2. PSS/N1336: Working effectively with others
3. PSS/N2501: Perform erection and commissioning of overhead lines, underground system and equipment
4. PSS/N2502: Perform erection and commissioning of railway substations

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution
Country	India
NSQF Level	4
Credits	NA

Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	10th Class
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	11/02/2019
Next Review Date	10/02/2023
NSQC Approval Date	22/08/2019
Version	1.0
Reference code on NQR	2019/POW/PSSC/03472
NQR Version	1.0

PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, assets and the environment

Scope

This unit/task covers the following:

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

PC1. • use protective clothing/equipment for specific tasks and work conditionsProtective Clothing • leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visorsequipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator

PC2. identify job-site hazards and possible causes of risks/accident at the workplaceHazards- electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions

PC3. follow procedures for safe working with electrical equipmentSafe Working Procedures: tag out/lock out, ptw (permit to work)

PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc

PC5. use standard safe working practices when working at heights, confined areas and trenches

PC6. test any electrical equipment and system using insulated testing devices before touching them

PC7. ensure positive isolation of electrical equipment & system as per the given standards

PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installedParameters: temperature, pressure, flow and current

PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.

PC10. use various methods of accident prevention in the work environmentMethods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors

PC11. identify and retrieve general health and safety equipment in the workplaceGeneral Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.

PC12. set up and use scaffolds, elevated platforms and ladders safelySet up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc

PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify themFaults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.

PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa

PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage

PC16. store flammable materials and machine lubricating oil safely and correctly

PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC18. use good housekeeping practices at all timesGood Housekee Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces

PC19. identify common hazard signs displayed in various areasVarious Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.

PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC21. use various appropriate fire extinguishers for different types of fires correctlyTypes of Fires: class a, b, c, d and e

PC22. use appropriate rescue techniques during fire hazard

PC23. maintain good housekeeping in order to prevent fire hazards

PC24. use a fire extinguisher correctly in case of fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC25. administer appropriate first aid to the victims for the relevant injuriesInjuries: bleeding, burns, choking, electric shock, poisoning etc.

PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments

PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments

PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services

PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergenciesEmergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work

PC30. free an affected person from electrocution safely as per the recommended procedure

PC31. move injured people and others to a safe place during emergencies

PC32. write an incident report or dictate a report to another person, and send report to the authorized personIncident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. names (and job titles if applicable), and where to find, all the people responsible for health and safety in a workplace
2. names and location of documents that refer to health and safety in the workplace
3. names and location of people responsible for health and safety in the workplace
4. documents that refer to health and safety in the workplacedocuments: fire notices, accident reports, safety instructions for equipment and procedures, company notices and documents, legal documents (e.g. government notices)
5. meaning of hazards and risks
6. health and safety hazards commonly present in the work environment and related precautions
7. possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
8. possible causes of risk and accident possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness); not taking safety precautions
9. methods of accident prevention
10. safe working practices when working with tools and machines
11. safe working practices while working at various hazardous sites
12. where to find all the general health and safety equipment in the workplace
13. various dangers associated with the use of electrical equipment
14. positive isolation of electrical equipment and system
15. safe handling and disposal of hazardous power plant wastes
16. use of emission and pollution control devices and measures taken to control pollution
17. various safety procedures and equipment used to work at heights, trenches and confined places
18. safe working practices specific to working with electrical equipment and system e.g. lock out/ tag out, ptw, etc.

19. preventative and remedial actions to be taken in the case of exposure to toxic materialstoxic materials: solvents, flux, lead exposure: ingested, contact with skin, inhaledpreventative action: ventilation, masks, protective clothing/ equipment); remedial action: immediate first aid, report to supervisor
20. importance of using protective clothing/equipment and other insulated work gear while handling electrical system and equipment
21. precautionary measures taken to prevent fire accident
22. various causes of firecauses of fires: heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.
23. correct technique(s) of using a fire extinguisher
24. different methods of extinguishing fire
25. different materials used for extinguishing firematerials: sand, water, foam, co2, dry powder
26. emergency rescue techniques applied during a fire hazard
27. various types of safety signs and what they mean
28. appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
29. rescue techniques applied during fire hazards
30. good housekeeping in order to prevent fire hazards
31. correct use of a fire extinguisher
32. how to free a person from electrocution
33. basic techniques of bandaging
34. perform artificial respiration and the cpr process

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write an accident/ incident report in local language and/or english
2. read and comprehend basic content to read labels, charts, signages
3. read and comprehend basic english to read manuals of operation
4. read an accident/ incident report in local language or english
5. question coworkers appropriately in order to clarify instructions and other issues
6. give clear instructions to coworkers and subordinates
7. remain congenial while discussing and debating issues with co-workers
8. follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
9. ask for, provide and receive required assistance where possible to ensure achievement of work-related objectives
10. follow organizational rule-based decision-making process
11. take decisions with systematic course of actions and/or response
12. plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
13. build customer relationships and use customer centric approach

14. think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
15. seek appropriate assistance from other sources to resolve problems
16. report problems that cannot be resolved to the appropriate authority
17. identify cause and effect relations in own area of work
18. work systematically and logically to resolve the issues, identify the cause and anticipate unexpected results
19. critically evaluate the organizational processes to improve them

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Follow workplace health and safety practices	**19**	**43**	**-**	**-**
PC1. • use protective clothing/equipment for specific tasks and work conditionsProtective Clothing • leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visorsequipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator	1	2	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplaceHazards-electrical hazards; sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; welding radiation; hazardous surfaces; hazardous substances; physical hazards possible causes of risk and accident: physical actions; not following instructions; inattention; sickness and incapacity; health; not taking safety precautions	1	2	-	-
PC3. follow procedures for safe working with electrical equipmentSafe Working Procedures: tag out/lock out, ptw (permit to work)	1	2	-	-
PC4. follow warning signs while working with electrical systems Warning Signs: danger, high voltage area, out of service, etc	1	2	-	-
PC5. use standard safe working practices when working at heights, confined areas and trenches	1	2	-	-
PC6. test any electrical equipment and system using insulated testing devices before touching them	1	2	-	-
PC7. ensure positive isolation of electrical equipment & system as per the given standards	1	1	-	-
PC8. identify any abnormalities in electrical equipment or system installed alarm and/or noticing parameters from gauge/ indicator installedParameters: temperature, pressure, flow and current	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. follow safe working practices while working in hazardous conditions to ensure the safety of self and others Safe Working Practices: using ppe; installing and reading safety signs; handle tools in the correct manner and store and maintain them properly; keep work area clear of clutter; while working with electricity take all electrical precautions; safe lifting and carrying practices; use equipment that is working properly and is well maintained; take due measures for safety while working at heights, etc. take due measures for safety while working in confined spaces or trenches, etc.	1	2	-	-
PC10. use various methods of accident prevention in the work environmentMethods of Accident Prevention: training in health and safety procedures; using health and safety procedures; use of equipment and working practices; safety notices, advice; instruction from colleagues and supervisors	1	2	-	-
PC11. identify and retrieve general health and safety equipment in the workplaceGeneral Health and Safety Equipment: fire extinguishers; first aid equipment; safety instruments and clothing etc.	1	2	-	-
PC12. set up and use scaffolds, elevated platforms and ladders safelySet up: firm/level base, clip/lash down, leaning at the correct angle, appropriate load as per capacity, etc	1	3	-	-
PC13. inspect scaffolds, elevated platforms and ladders for faults and rectify themFaults: corrosion of metal components, deterioration, splits and cracks in timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc.	1	3	-	-
PC14. lift, carry and transport heavy objects & tools safely using correct procedures from storage to the workplace and vice versa	-	3	-	-
PC15. inspect power plant and its equipment routinely for any signs of oil, water and steam leakage	1	3	-	-
PC16. store flammable materials and machine lubricating oil safely and correctly	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	2	-	-
PC18. use good housekeeping practices at all timesGood Housekee Practices: clean/tidy work area, removal/disposal of waste material, protect surfaces	1	2	-	-
PC19. identify common hazard signs displayed in various areasVarious Areas: chemical containers; equipment; packages; inside buildings; open areas and public spaces, etc.	1	2	-	-
PC20. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	2	-	-
Follow fire safety practices	**3**	**8**	**-**	**-**
PC21. use various appropriate fire extinguishers for different types of fires correctlyTypes of Fires: class a, b, c, d and e	1	2	-	-
PC22. use appropriate rescue techniques during fire hazard	1	2	-	-
PC23. maintain good housekeeping in order to prevent fire hazards	1	2	-	-
PC24. use a fire extinguisher correctly in case of fire	-	2	-	-
Follow emergencies, rescue and first-aid procedures	**8**	**19**	**-**	**-**
PC25. administer appropriate first aid to the victims for the relevant injuriesInjuries: bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC26. act promptly and appropriately to an accident situation or medical emergency in real or simulated environments	1	2	-	-
PC27. perform and organise loss minimization or rescue activity during an accident in real or simulated environments	1	4	-	-
PC28. administer basic first aid including cpr to the victims of heart or cardiac arrest due to electric shock, before the arrival of emergency services	1	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. demonstrate basic emergency procedures during emergency mock drills for better understanding of procedures applied in emergenciesEmergency Procedures: raising alarm, safe/efficient evacuation, correct ways of escape, correct assembly point, roll call, correct return to work	1	2	-	-
PC30. free an affected person from electrocution safely as per the recommended procedure	1	2	-	-
PC31. move injured people and others to a safe place during emergencies	-	3	-	-
PC32. write an incident report or dictate a report to another person, and send report to the authorized personIncident Report Includes Details of: name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified	2	2	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Distribution, Transmission, Generation, Distribution Downstream
Occupation	Operation & Maintenance - Generation Supervision - Distribution Erection, Installation and Commissioning - DistributionSurveying - TransmissionOperation & Maintenance - Transmission
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	11/02/2023
NSQC Clearance Date	22/08/2019

PSS/N1336: Working effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behavior and interactions with others at the workplace. These cover areas such as communication etiquette,discipline, listening, handling conflict and grievances.

Elements and Performance Criteria

working with others

To be competent, the user/individual on the job must be able to:

PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required

PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt

PC3. give information to others clearly, at a pace and in a manner that helps them to understand

PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible

PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks

PC6. display appropriate communication etiquette while working

PC7. display active listening skills while interacting with others at work

PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism

PC9. demonstrate responsible and disciplined behaviors at the workplace

PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. relevant people and their responsibilities within the work area

KU4. eescalation matrix and procedures for reporting work and employment related issues

KU5. various categories of people that one is required to communicate and co-ordinate with in the organization

KU6. importance of effective communication in the workplace

KU7. importance of teamwork in organizational and individual success

KU8. various components of effective communication

KU9. key elements of active listening

KU10. value and importance of active listening and assertive communication

KU11. barriers to effective communication

KU12. importance of tone and pitch in effective communication

KU13. importance of avoiding casual expletives and unpleasant terms while communicating professional circles

KU14. how poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer

KU15. importance of ethics for professional success

KU16. importance of discipline for professional success

KU17. what constitutes disciplined behavior for a working professional

KU18. common reasons for interpersonal conflict

KU19. importance of developing effective working relationships for professional success

KU20. expressing and addressing grievances appropriately and effectively

KU21. importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. note the information communicated by the officer incharge

GS2. note down observations (if any) related to the operation/maintenance

GS3. read and interpret the process required for different types of manuals

GS4. read and interpret the flowchart of all parts of an assembly

GS5. read manuals and documents to understand the product-details & how they can be used

GS6. discuss task lists, schedules and activities with the colleague/supervisor

GS7. effectively communicate with the team members

GS8. attentively listen and comprehend the information given by the colleague/supervisor/contractor

GS9. communicate clearly with the colleague on the issues faced during query/fault

GS10. follow colleague/contractor rule-based decision making process

GS11. take decisions with systematic course of actions and/or response

GS12. planning and organization of tasks to meet deadlines

GS13. build customer relationships and use customer centric approach

GS14. seek and comprehend operation related inputs for clarification find ways of modifying difficult operating stages to make it operation friendly

GS15. work systematically and logically to resolve the issues and identify causation and anticipate unexpected results. Quick approach and solution towards faults repairing

GS16. critically evaluate operation parameters in relation to system normality develop a holistic and comprehensive profile of grid station on segregated discrete processes

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
working with others	**30**	**70**	**-**	**-**
PC1. accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	3	7	-	-
PC2. accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	3	7	-	-
PC3. give information to others clearly, at a pace and in a manner that helps them to understand	3	7	-	-
PC4. display helpful behavior by assisting others in performing tasks in a positive manner, where required and possible	3	7	-	-
PC5. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	7	-	-
PC6. display appropriate communication etiquette while working	3	7	-	-
PC7. display active listening skills while interacting with others at work	3	7	-	-
PC8. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	3	7	-	-
PC9. demonstrate responsible and disciplined behaviors at the workplace	3	7	-	-
PC10. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	7	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	TBD
Version	1.0
Last Reviewed Date	31/03/2022
Next Review Date	31/03/2025
NSQC Clearance Date	01/03/2023

PSS/N2501: Perform erection and commissioning of overhead lines, underground system and equipment

Description

This unit covers the railway track electrification activities such as testing of soil, erection of mast, stringing of overhead lines, installation of OHE, testing and commissioning of lines, OHE, laying of underground cables etc. as per the RDSO guidelines/ procedures

Scope

This unit/ task covers the following:

- Erect the mast
- String lines, underground cables and OHE
- Test and commission lines, underground cables and OHE
- Perform energization of overhead lines

Elements and Performance Criteria

Erect the mast

To be competent, the user/individual on the job must be able to:

PC1. organize for PTW (permit to work) in both Power Block and Traffic Block from concerned authorities

PC2. perform soil test as per Research Designs and Standards Organisation (RDSO) standards

PC3. analyse the data related to the type of civil foundation required based on the soil test results

PC4. monitor and inspect the foundation work building process according to RDSO standards

PC5. read and interpret the layout plan for erection of mast

PC6. select a suitable mast depending on the loading condition, wind load etc.

PC7. erect the mast as per the standard procedure laid in RDSO

String lines, underground cables and OHE

To be competent, the user/individual on the job must be able to:

PC8. identify various overhead equipment used in railway track electrification Various overhead equipment: Stay tube insulator, tubular stay arm, standard / large bracket, register arm, steady arm, bracket insulator, catenary suspension bracket, register arm hook, drop bracket, anti-wind clamp etc.

PC9. mark the anchor points based on the overhead equipment (OHE) used

PC10. select a suitable anti-creep point

PC11. fix the overhead equipment and carry out stringing of lines

PC12. insert droppers as per the required schedule

PC13. fix the Auto Tensioning Device (ATD) and adjust the tension of the overhead lines as per the RDSO standards

PC14. erect isolators as per the drawing/ layout

PC15. erect section insulators on loop lines and cross overs

PC16. identify and erect neutral section at suitable location and also differentiate between various types of neutral sections

PC17. select turn out locations and crossing

PC18. carry out earthing of every mast and other OHE

PC19. provide bonding of masts according to standard norms

PC20. obtain necessary clearance from Power Distribution Utilities to increase the pole height or to underground the distribution lines in form of cables to avoid interference of distribution lines and railway lines

PC21. perform cable jointing

PC22. erect the emergency mast

Test and commission lines, underground cables and OHE

To be competent, the user/individual on the job must be able to:

PC23. test all the Tools and Tackles (T and P) for their safe and efficient working

PC24. inspect the lines and OHE thoroughly as per the RDSO standards

PC25. check the line height suitability for transportation of ODC (Over Dimensional Consignment)

PC26. check or measure the stagger implantation as per requirement

PC27. check the tension on the wire using a dynamometer

PC28. check that the height of the contact wire is as per standards

PC29. ensure that the versine angle is in adherence with RDSO standards

PC30. ensure that the maximum blow off is under permissible limits

PC31. ensure that the electrical clearances are as per the RDSO standards

PC32. check that the cable joints are done as per standards

PC33. ensure that the cable insulation is as per the manufacturers specifications

Perform energization of overhead lines

To be competent, the user/individual on the job must be able to:

PC34. obtain necessary safety clearance regarding overhead line energization from the concerned authority

PC35. energize the overhead lines from the respective feeder

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the company
2. relevant health and safety requirements applicable in the work place
3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
4. reporting structure, inter-dependent functions, lines and procedures in the work area
5. principles of electricity
6. principles and practices of electrical safety

7. common electricity terminology and correct interpretation of the same Terminology: e.g. Current, Voltage, Resistance, Inductance, Capacitance, Kilovolt ampere (kVA), Kilowatt (kW), active power, Kilowatt hour (kWh) (unit of electric consumption), Power factor
8. different types of material and accessories used in railway track electrification Materials and accessories: e.g. mast fitting for various types of insulators such as ST (Suspension type) insulator, stay tube insulator, tubular stay arm, standard / large bracket, register arm, steady arm, bracket insulator, catenary suspension bracket, register arm hook, drop bracket, anti-wind clamp, inclined dropper, suspension clamp, contact wire swivel clip, tubular stay sleeve, tubular stay adjuster, tube cap
9. tools and equipment used in testing, erection and commissioning Tools: e.g. Plier, Screwdriver, Wrench set, Hammer, Drilling machine, Hacksaw / cutting tools, Measuring tape, Pulleys (Force Pulley with sling), Tommy bar, Crimping machine, Round / flat file, Earth rod (discharge rod), torue wrench etc
10. use of soil test results as per RDSO standards
11. use of civil foundation standards as per RDSO
12. applicable limit of tension on overhead line
13. use and function of isolators, interrupters
14. types of Neutral Sections
15. types of turnout and crossing
16. types of bonding used in masts in railway track electrification
17. types and sizes of conductors and cables
18. difference between various types of cables
19. various cable jointing techniques
20. procedure for erection of emergency mas
21. importance of leaving the work area and equipment in a safe and clean condition on completion of the repair and maintenance activities
22. importance of reporting problems in a timely manner
23. personal protective equipment (PPE) and clothing that must be worn during the inspection, repair and maintenance activity and from where can it be obtained PPE: e.g. Safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), Zola, safety rope
24. how to undertake basic numerical computations and calculations Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
25. how to identify various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Compound shapes: involving squares, rectangles, triangles, circles, semicircles, quadrants of a circle Solid shapes: cube, rectangular prism, cylinder Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity
26. how to use metric systems of measurement
27. how to participate in on-the-job and other learning, training and development interventions and assessments
28. how to maintain current knowledge of application standards, legislation, codes of practice and product/ process developments

Generic Skills (GS)

User/individual on the job needs to know how to:

1. fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
2. read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language
3. convey and share technical information clearly using appropriate language
4. clarify task related information with appropriate personnel or technical advise
5. communicate and cooperate with colleagues in the team for better results
6. seek assistance from fellow team members
7. follow organizational rule-based decision-making process
8. plan, prioritize and sequence work operations as per job requirements
9. organize and analyze information relevant to work
10. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
11. work in a team in order to achieve better results
12. identify and clarify work roles within a team

. avoid and manage distractions to be disciplined at work

14. manage own time for achieving better results
15. identify and express new ideas and initiatives to others
16. participate in improvement procedures including process, quality and internal/ external customer/ supplier relationships
17. exercise restraint while expressing dissent and during conflict situations
18. identify problems with work planning, procedures, output and behavior and their implications
19. prioritize and plan for problem solving
20. communicate problems appropriately to others
21. identify sources of information and support for problem solving
22. modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
23. improve and modify own work practices
24. analyse the problems in the equipment to identify root causes
25. collect information and technical data and define process for equipment testing
26. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
27. develop holistic and comprehensive profile of products based on segregated discrete process stages
28. apply, analyze and evaluate the information gathered from observation,
29. experience, reasoning, or communication, as a guide to thought and action
30. participate in on-the-job and other learning, training and development

31. interventions and assessments
32. clarify task related information with appropriate authority
33. seek to improve and modify own work practices
34. maintain current knowledge of application standards, legislation, codes of practice and product/process developments

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Erect the mast	**8.5**	**16.5**	**-**	**-**
PC1. organize for PTW (permit to work) in both Power Block and Traffic Block from concerned authorities	1	3	-	-
PC2. perform soil test as per Research Designs and Standards Organisation (RDSO) standards	2	3	-	-
PC3. analyse the data related to the type of civil foundation required based on the soil test results	1	3	-	-
PC4. monitor and inspect the foundation work building process according to RDSO standards	1	2	-	-
PC5. read and interpret the layout plan for erection of mast	2	2	-	-
PC6. select a suitable mast depending on the loading condition, wind load etc.	0.5	1.5	-	-
PC7. erect the mast as per the standard procedure laid in RDSO	1	2	-	-
String lines, underground cables and OHE	**13.5**	**30.5**	**-**	**-**
PC8. identify various overhead equipment used in railway track electrification Various overhead equipment: Stay tube insulator, tubular stay arm, standard / large bracket, register arm, steady arm, bracket insulator, catenary suspension bracket, register arm hook, drop bracket, anti-wind clamp etc.	0.5	1.5	-	-
PC9. mark the anchor points based on the overhead equipment (OHE) used	1	1	-	-
PC10. select a suitable anti-creep point	1	2	-	-
PC11. fix the overhead equipment and carry out stringing of lines	1	2	-	-
PC12. insert droppers as per the required schedule	1	3	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. fix the Auto Tensioning Device (ATD) and adjust the tension of the overhead lines as per the RDSO standards	1	2	-	-
PC14. erect isolators as per the drawing/ layout	1	2	-	-
PC15. erect section insulators on loop lines and cross overs	1	2	-	-
PC16. identify and erect neutral section at suitable location and also differentiate between various types of neutral sections	1	2	-	-
PC17. select turn out locations and crossing	1	2	-	-
PC18. carry out earthing of every mast and other OHE	1	2	-	-
PC19. provide bonding of masts according to standard norms	1	2	-	-
PC20. obtain necessary clearance from Power Distribution Utilities to increase the pole height or to underground the distribution lines in form of cables to avoid interference of distribution lines and railway lines	1	2	-	-
PC21. perform cable jointing	-	3	-	-
PC22. erect the emergency mast	1	2	-	-
Test and commission lines, underground cables and OHE	**7**	**19**	**-**	**-**
PC23. test all the Tools and Tackles (T and P) for their safe and efficient working	2	2	-	-
PC24. inspect the lines and OHE thoroughly as per the RDSO standards	1	3	-	-
PC25. check the line height suitability for transportation of ODC (Over Dimensional Consignment)	0.5	1.5	-	-
PC26. check or measure the stagger implantation as per requirement	0.5	1.5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. check the tension on the wire using a dynamometer	0.5	1.5	-	-
PC28. check that the height of the contact wire is as per standards	0.5	1.5	-	-
PC29. ensure that the versine angle is in adherence with RDSO standards	0.5	1.5	-	-
PC30. ensure that the maximum blow off is under permissible limits	-	2	-	-
PC31. ensure that the electrical clearances are as per the RDSO standards	0.5	1.5	-	-
PC32. check that the cable joints are done as per standards	0.5	1.5	-	-
PC33. ensure that the cable insulation is as per the manufacturers specifications	0.5	1.5	-	-
Perform energization of overhead lines	**1**	**4**	**-**	**-**
PC34. obtain necessary safety clearance regarding overhead line energization from the concerned authority	1	2	-	-
PC35. energize the overhead lines from the respective feeder	-	2	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2501
NOS Name	Perform erection and commissioning of overhead lines, underground system and equipment
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning- Distribution
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	10/02/2023
NSQC Clearance Date	22/08/2019

PSS/N2502: Perform erection and commissioning of railway substations

Description

This unit covers the competencies required by technicians for erection and commissioning of railway substation. This unit also covers the skills and competencies needed for erection and commissioning of other equipment associated with railway substation

Scope

This unit/task covers the following:

- Perform erection and commissioning of railway substations

Elements and Performance Criteria

Perform erection and commissioning of railway substations

To be competent, the user/individual on the job must be able to:

PC1. read and interpret layout plan of the substation
PC2. ensure that the unloading of the traction transformer at the site is done as per norms
PC3. check for any damage to the traction transformer and its components during transportation
PC4. assemble the accessories of the traction transformer at the site
PC5. carry out pipe and plate earthing to make earth connection and earth mat
PC6. ensure that each earth pit be marked with date and next due date for testing of the earth resistance
PC7. ensure the availability of double earth connection to each equipment
PC8. ensure installation of all substation equipment such as CT (current transformer), CB (circuit breaker), LA (lightening arrestor), PT (potential transformer) etc. is done as per norms
PC9. ensure that the voltage level of equipment is as per the specified rating
PC10. ensure that the equipment with appropriate voltage level has been fixed appropriately as per drawings
PC11. ensure proper connections being made to OHE through feeding post
PC12. ensure erection of SP (switching posts) and SSP (Sub-sectioning and Paralleling post) as per standards at appropriate location
PC13. erect and install interrupters on SP and SSP
PC14. erect the auxiliary transformer for signaling purpose as per specifications and guidelines
PC15. check that the painting of all equipment is done as per standards
PC16. ensure that the gravel laid in the substation yard is as per specified norms
PC17. ensure availability of proper lighting arrangement at the substation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the
2. company relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
5. escalation matrix and procedures for reporting work and employment related issues
6. principles of electricity Power system: how power flows, generation, transmission and distribution number of bays, number of incoming and outgoing feeders, load management through single or double bus, substation network, ring system, back feed etc
7. line components towers, poles, single circuit, double circuit, overhead, underground conductors and cables
8. function of transformer accessories
9. gantry structure, structure lay out, types of porcelain insulators, overhead conductors, clamps used in station
10. outdoor and indoor equipments like power transformer, circuit breaker, Isolator, LA, CT, PT, and CVT
11. lightening arrestors (LA) functioning
12. difference between various switching stations
13. types of earthing used in grid station, its significance, why earth connection with each equipments, measurement of earth resistance and earthing switch.
14. complete tools, tackles and safety gadgets required in grid station erection and commissioning
15. how to keep records of all equipment like name plate, pre- commission test report and manuals
16. hazards associated with carrying out substation construction and installation process and how they can be minimized
17. how to undertake basic numerical computations and calculations Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
18. how to use metric systems of measurement Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity
19. how to participate in on-the-job and other learning, training and development interventions and assessments
20. how to maintain current knowledge of application standards, legislation,codes of practice and product/ process developments

Generic Skills (GS)

User/individual on the job needs to know how to:

1. fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
2. read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language
3. convey and share technical information clearly using appropriate language

4. clarify task related information with appropriate personnel or technical advise
5. communicate and cooperate with colleagues in the team for better results
6. seek assistance from fellow team members
7. follow organizational rule-based decision-making process
8. plan, prioritize and sequence work operations as per job requirements
9. organize and analyze information relevant to work
10. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
11. work in a team in order to achieve better results
12. identify and clarify work roles within a team
13. avoid and manage distractions to be disciplined at work
14. manage own time for achieving better results
15. identify and express new ideas and initiatives to others
16. participate in improvement procedures including process, quality and internal/ external customer/ supplier relationships
17. exercise restraint while expressing dissent and during conflict situations
18. identify problems with work planning, procedures, output and behavior and their implications
19. prioritize and plan for problem solving
20. communicate problems appropriately to others
21. identify sources of information and support for problem solving
22. modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
23. improve and modify own work practices
24. analyse the problems in the equipment to identify root causes
25. collect information and technical data and define process for equipment testing
26. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
27. develop holistic and comprehensive profile of products based on segregated discrete process stages
28. apply, analyze and evaluate the information gathered from observation,
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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Perform erection and commissioning of railway substations	**30**	**70**	**-**	**-**
PC1. read and interpret layout plan of the substation	2	4	-	-
PC2. ensure that the unloading of the traction transformer at the site is done as per norms	1	3	-	-
PC3. check for any damage to the traction transformer and its components during transportation	1	3	-	-
PC4. assemble the accessories of the traction transformer at the site	2	6	-	-
PC5. carry out pipe and plate earthing to make earth connection and earth mat	2	6	-	-
PC6. ensure that each earth pit be marked with date and next due date for testing of the earth resistance	2	4	-	-
PC7. ensure the availability of double earth connection to each equipment	2	3	-	-
PC8. ensure installation of all substation equipment such as CT (current transformer), CB (circuit breaker), LA (lightening arrestor), PT (potential transformer) etc. is done as per norms	2	4	-	-
PC9. ensure that the voltage level of equipment is as per the specified rating	2	3	-	-
PC10. ensure that the equipment with appropriate voltage level has been fixed appropriately as per drawings	2	4	-	-
PC11. ensure proper connections being made to OHE through feeding post	2	4	-	-
PC12. ensure erection of SP (switching posts) and SSP (Sub-sectioning and Paralleling post) as per standards at appropriate location	2	4	-	-
PC13. erect and install interrupters on SP and SSP	3	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. erect the auxiliary transformer for signaling purpose as per specifications and guidelines	2	5	-	-
PC15. check that the painting of all equipment is done as per standards	1	4	-	-
PC16. ensure that the gravel laid in the substation yard is as per specified norms	1	4	-	-
PC17. ensure availability of proper lighting arrangement at the substation	1	4	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2502
NOS Name	Perform erection and commissioning of railway substations
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning- Distribution
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	11/02/2019
Next Review Date	10/02/2023
NSQC Clearance Date	22/08/2019

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1.Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	30	70	-	-	100	15
PSS/N1336.Working effectively with others	30	70	-	-	100	5
PSS/N2501.Perform erection and commissioning of overhead lines, underground system and equipment	30	70	-	-	100	40
PSS/N2502.Perform erection and commissioning of railway substations	30	70	-	-	100	40
Total	**120**	**280**	**-**	**-**	**400**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
RDSO	Research Design And Standard Organization
OHE	Over Head Equipment
ATD	Auto Tensioning Device
PTFE	Poly Tetra Fluoro Ethene
ODC	Over Dimensional Consignment
EIG	Electrical Inspector General
CRS	Commission Of Railway Safety
TSS	Traction Sub-Station
SP	Sectioning And Paralleling Post
SSP	Sub-Sectioning And Paralleling Post
YS	Yard Supply Post
MH	Material Handling
CT	Current Transformer
PT	Potential Transformer
CVT	Capacitive Voltage Transformer
LA	Lightening Arrestor
CB	Circuit Breaker

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.